

OPS-RB-005 - Deployment Rollback Procedure

Operational Procedure | Operations

Document ID	OPS-RB-005	Seller ID	Organization-wide
Department	Operations	Classification	INTERNAL
Owner	Neha Gupta	Created Date	2026-01-29
Related Scenario ID	SC-010	Document Type	Operational Procedure

1. Purpose

This procedure defines how a release is withdrawn from production when it is causing harm, and how the decision between rolling back and rolling forward is made.

2. Scope

Applies to any production release of a platform service. Applies to configuration changes deployed through the release pipeline. Applies to schema migrations, with the additional constraints in section 7.

Does not apply to emergency configuration changes applied outside the pipeline, which follow the Incident Escalation Procedure.

3. Preconditions

1. An incident is open, or is opened as part of invoking this procedure.
2. The harm is observed in production, not suspected.
3. The release under suspicion is identified, with its deployment start time.
4. A named decision owner is present: the Engineering incident owner.

Correlation with a deployment time is sufficient to invoke this procedure. Confirmed causation is not required and waiting for it is usually the wrong choice.

4. Responsibilities

Role	Responsibility
Engineering incident owner	Rollback decision and execution
Operations	Verification that service and data returned to correct state
Support	Seller communication
Release owner	Root cause analysis and the corrected release

5. Rollback Versus Roll Forward

This is the only significant decision in the procedure.

Choose rollback when	Choose roll forward when
The cause is not yet understood	The cause is understood and the fix is small
Harm is accruing now	Harm is bounded and not growing
The previous version is known good	The previous version has its own known defect

Choose rollback when	Choose roll forward when
No schema migration blocks reversion	A schema migration cannot be reversed safely
The failure mode disappears once the fleet is homogeneous	The failure exists in both versions

Default to rollback. Understanding a defect takes longer than reverting one, and a rollback returns the system to a known state. Roll forward only where rollback is unsafe or where the fix is genuinely trivial and understood.

In August 2026 a checkout release caused token validation failures. The cause was not understood at the point of decision, harm was accruing on a shared revenue path, and the failure mode was known to disappear once all instances ran the same version. Rollback was initiated 4 minutes after the decision point and completed in 34 minutes. Root cause was confirmed five hours later, comfortably after service was restored.

6. Procedure

Step 1 - Halt the Deployment

Stop the rollout before deciding anything else. A partially deployed release that is causing harm will cause more harm as it proceeds, and a mixed fleet is often the failure condition itself.

Step 2 - Decide

1. Apply section 5.
2. Record the decision and its reasoning in the incident record.
3. Name the executor.

Target: decision within 10 minutes of the deployment being halted.

Step 3 - Execute Rollback

1. Revert to the last known good version, instance by instance.
2. Monitor the harm metric continuously during reversion. It should improve proportionally as instances revert.
3. If the metric does not improve as instances revert, **stop**. The release is probably not the cause and reverting further adds risk without benefit.
4. Confirm the fleet is homogeneous at the reverted version.

Step 4 - Verify Service

Check	Acceptance
Harm metric	Returned to pre-deployment baseline
Error rates	At baseline across the affected service
Latency	At baseline
Fleet version consistency	All instances on the same version
Dependent services	No residual errors

Step 5 - Verify Data

A rollback restores code. It does not restore data written by the defective version.

- 1. Determine whether the defective version wrote incorrect data.
- 2. If it did, hand off to the Service Recovery Procedure with a named recovery owner.
- 3. Do not close the incident on service restoration alone.

Step 6 - Communicate

- 1. Confirm to Support that service is restored so that sellers can be told.
- 2. Where a seller reported the issue, confirm resolution against their original report rather than against internal metrics.

Step 7 - Corrected Release

The release owner produces the corrected version. It must include:

- 1. The fix for the defect.
- 2. A control that prevents the same class of defect. For a data contract change, this normally means a transition mechanism such as dual-read, or an explicit version field, rather than a one-way change.
- 3. Test coverage for the condition that was missed.

Re-release only after the corrected version has been validated against the specific failure condition, not merely against the normal path.

7. Schema Migrations

Schema migrations constrain rollback and require additional care.

- 1. A migration must be backward compatible with the previous application version for at least one release cycle.
- 2. A migration that cannot be reversed must be identified as such before deployment, and the release must be deployable and revertible independently of it.
- 3. Never combine an irreversible migration with an application change in the same deployment.
- 4. A migration that alters a uniqueness guarantee, a key, or a constraint on financial data requires financial review before deployment, not only engineering review.

Requirement 4 exists because a deduplication key change during a schema migration in early 2026 removed a uniqueness guarantee on settlement ledger reversal entries and produced an incorrect payout. The change was reviewed as a schema change; its effect on financial correctness was not examined.

8. Decision Points

Decision	Criterion	Owner
Halt the deployment	Harm observed and correlated with the rollout	Engineering
Rollback or roll forward	Section 5	Engineering incident owner

Decision	Criterion	Owner
Stop a rollback mid-execution	Harm metric not improving with reversion	Engineering incident owner
Hand off to data recovery	Defective version wrote incorrect data	Operations
Re-release	Corrected version validated against the failure condition	Release owner

9. Exceptions

A release may be left in production despite observed harm where the harm is below an agreed threshold, the fix is already in flight, and a named owner accepts the residual. Record the acceptance.

A rollback may be skipped during a change freeze only if the alternative is demonstrably lower risk. A change freeze is not a reason to leave a harmful release in production.

10. Escalation

Situation	Escalate to	Timeframe
Rollback fails or cannot complete	Engineering lead and Operations	Immediately
Harm metric does not improve on reversion	Engineering incident owner	Immediately
Rollback blocked by an irreversible migration	Engineering lead and Operations	Immediately
Data written incorrectly by the defective version	Operations, for recovery	Before incident closure
Decision not reached within 10 minutes	Engineering lead	On breach

11. Post-Incident Actions

1. Record time from deployment start to harm detection, and from detection to mitigation.
2. Determine whether the defect was detectable before production. Where the gap is a missing pipeline stage, such as mixed-version testing, raise it as a preventive action.
3. Check whether the relevant alert threshold was capable of detecting the degradation. Thresholds tuned for outages will not catch partial degradations.
4. Confirm the release notes declared any cross-instance or cross-service contract change.

12. Revision History

Version	Date	Change
1.0	2024-11-05	Initial procedure
1.1	2025-06-17	Schema migration constraints added

Version	Date	Change
2.0	2026-01-29	Default to rollback; decision target of 10 minutes
2.1	2026-04-30	Financial review required for migrations affecting financial constraints
2.2	2026-08-18	Data verification separated from service verification